

Transformations — every rule on one page

Each rule goes up on the wall the day the class discovers it — not before.

RIGID TRANSFORMATIONS — the image is CONGRUENT to the original

Transformation	Rule	What to remember
Translation { h, k }	$(x, y) \rightarrow (x + h, y + k)$	Read the signs. { 6, -3 } is right 6 and DOWN 3.
Rotation 90° ccw	$(x, y) \rightarrow (-y, x)$	Swap, then negate the new first coordinate.
Rotation 180°	$(x, y) \rightarrow (-x, -y)$	Negate both. Do not swap.
Rotation 270° ccw	$(x, y) \rightarrow (y, -x)$	Same as 90° clockwise.
Reflect over the x-axis	$(x, y) \rightarrow (x, -y)$	x stays. y flips.
Reflect over the y-axis	$(x, y) \rightarrow (-x, y)$	y stays. x flips.
Reflect over $y = x$	$(x, y) \rightarrow (y, x)$	SWAP them. Do not negate.
Reflect over $y = -x$	$(x, y) \rightarrow (-y, -x)$	Swap AND negate both.
Reflect over $x = a$ or $y = b$	no rule — count squares	Same number of squares on the other side of the line.

NOT RIGID — the image is SIMILAR but not congruent

Transformation	Rule	What to remember
Dilation, center the origin	$(x, y) \rightarrow (kx, ky)$	Multiply BOTH coordinates. $k > 1$ bigger, $0 < k < 1$ smaller.
Perimeter under a dilation	multiplied by k	Lengths scale by k.
Area under a dilation	multiplied by k^2	$k = 3$ makes the area 9 times as large, not 3.

SEQUENCES (COMPOSITIONS)

First image	A' — read “A prime”
Second image	A'' — read “A double prime”
The one rule that matters	The second transformation acts on the IMAGE, never on the original.
Does order matter?	Usually yes. Two translations are the exception.
x-axis then y-axis	the same as one rotation of 180° about the origin
Two reflections over parallel lines	a translation of twice the distance between them

congruent same size and shape. **similar** same shape, different size. **rigid** every length unchanged — translations, rotations and reflections.